

Scene Representations for Robotic Spatial Perception

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Abstract

The ability of a robot to build a persistent, accurate, and actionable model of its surroundings through sensor data in a timely manner is crucial for autonomous operation. While representing the world as a point cloud might be sufficient for localization, denser scene representations are required for obstacle avoidance. On the other hand, higher-level semantic information is often crucial for breaking down the necessary steps to autonomously complete a complex task, such as cooking. So the looming question is, What is a suitable scene representation for the robotic task at hand? This survey provides a comprehensive review of key approaches and frameworks driving progress in the field of robotic spatial perception, with a particular focus on the historical evolution and current trends in representation. By categorizing scene modeling techniques into three main types—metric, metric–semantic, and metric–semantic–topological—we discuss how spatial perception frameworks are transitioning from building purely geometric models of the world to more advanced data structures incorporating higher-level concepts, such as the notion of object instances and places. Special emphasis is placed on approaches for real-time simultaneous localization and mapping, their integration with deep learning for enhanced robustness and scene understanding, and their ability to handle scene dynamicity as some of the hottest topics of interest driving robotics research today. We conclude with a discussion of ongoing challenges and future research directions in the quest to develop robust and scalable spatial perception systems suitable for long-term autonomy.

1. INTRODUCTION

Equipping mobile robots with the ability to perceive and understand their surroundings is crucial for enabling autonomous operation and expanding their range of applications in complex, real-world scenarios. Central to this capability is the process of capturing key aspects of a robot's workspace in a map, effectively constructing a spatial representation of the environment, based on the data collected by its onboard sensors. Such a representation serves as the cornerstone of the robot's spatial awareness, allowing for the estimation of its own state and strongly influencing its ability to navigate, avoid obstacles, and interact intelligently and safely with objects and humans.

The field of robotic perception has witnessed significant progress throughout the years, propelled by advances in sensor technology, computational power, and algorithm development. Irrespective of the choice of sensors, which has a direct effect on the type of information that is captured, the presence of noise in every sensor measurement has long been a core subject of research for effective mapping. In addition, the particular need for robots to gather spatial information in a timely manner has placed the focus on the development of fast and efficient data management schemes, favoring real-time processing over the offline methods commonly used in photogrammetry and computer vision applications. Initial efforts to address these challenges were already made in the 1980s and early 1990s, relying primarily on basic grid-based representations that were limited in both accuracy and scalability (1). However, the introduction of probabilistic techniques, such as Kalman filters and expectation-maximization approaches, soon enabled robots to more rigorously handle pose uncertainty and noise in sensor data. These advancements laid the foundation for the mathematical formulation of the simultaneous localization and mapping (SLAM) problem (2, 3), which involves concurrently estimating the scene structure and localizing the robot relative to its growing map. Since the emergence of the first vision-based SLAM frameworks in the early 2000s (4), the field has continued to evolve progressively, and new approaches leveraging advancements in optimization algorithms have emerged, improving the trade-offs among accuracy, efficiency, and robustness within the estimation processes. In parallel, the rising popularity and portability of 3D sensors, such as lidar scanners and RGB-D cameras, have enabled the leap toward denser and more detailed 3D scene representations.

For the past decade or so, the adoption of deep learning has further revolutionized the landscape of robotic perception by providing the ability to process, analyze, and extract useful information from vast bodies of raw sensor data far more effectively than ever before. Neural networks (NNs) have been employed to enhance the identification of salient regions in the robot's workspace, increase robustness to environmental changes, and integrate semantic understanding into maps. These advances have transformed SLAM from a purely geometric problem into a more holistic framework for spatial perception, allowing robots not only to map the environment with greater accuracy and reliability but also to understand and interact with their surroundings in a more meaningful way. Furthermore, in recent years, neural scene representations have introduced a new paradigm for the way robots perceive and model their surroundings. By leveraging the power of NNs to encode the geometry, appearance, and, optionally, semantics of a scene into compact latent spaces, perception frameworks can now learn directly from raw sensory inputs and capture intricate details in a more scalable and flexible fashion compared with traditional mapping methods relying on explicit geometric primitives, such as voxels (i.e., volumetric elements equivalent to 3D pixels) or meshes.

Despite impressive progress so far, robotic spatial perception continues to face several challenges. While there exist well-accepted, converged solutions for simple and isolated robotic tasks, such as localization in static environments or path planning with known poses and maps, a significant gap remains in methods capable of addressing the dynamic, unpredictable, and complex

nature of real-world scenes. Additionally, computational efficiency becomes a critical issue as the need to execute sophisticated perception algorithms within the constraints of onboard processing capabilities rises. At the same time, ensuring longer-term robustness and reliability in diverse large-scale environments is imperative to enable sustained operation in real missions. Addressing these issues requires further investigation into suitable scene representations, as these provide the foundational data structures necessary for accurate perception, efficient computation, and reliable decision-making in varied and demanding scenarios.

Inspired by the progress and the ongoing challenges, this article explores the latest advancements in scene representation for robotic spatial perception. Studying the most important works in the field over the past decade in a comprehensive overview of the current state of the art, we draw insights highlighting key approaches and frameworks that have driven progress and show promise. Emanating from this study is a discussion of remaining open questions and promising future directions toward the goal of equipping robots with the advanced spatial awareness necessary to perform autonomously in challenging real-world missions.

2. RELATION TO PRIOR SURVEYS AND SCOPE

Research in robotic spatial perception has a long history, and many survey papers addressing core topics such as SLAM and high-level scene understanding have been published over the years. In the 2000s, Thrun (1) provided a detailed discussion on early robot mapping techniques, while tutorials by Durrant-Whyte and Bailey (2, 3) and Grisetti et al. (5) outlined the probabilistic nature of the SLAM problem and its graph-based formulation, respectively, which still constitute the foundation of many state-of-the-art frameworks. As the field began to mature, the topics of robustness, scalability, and representation (both metric and semantic) started to raise concerns within the community and were extensively discussed in a survey by Cadena et al. (6), which marked the beginning of what was termed the Robust-Perception Age. By that time (i.e., the mid-2010s), the algorithmic constructs of the SLAM problem had become established, and the availability of high-quality, open-source factor graph solvers facilitated the development of the first modern visual SLAM implementations, most of which were described in a survey by Taketomi et al. (7). However, the surge of deep learning opened up exciting new possibilities for robotic perception, causing a shift of focus toward increasing the representational richness necessary to enable robust and intelligent autonomy and sparking enormous interest in the field.

Numerous surveys studying the latest wave of SLAM and spatial perception systems have been published in recent years. An extensive review by Garg et al. (8) focused on the use of semantics for robot mapping, perception, and interaction, while Rosen et al. (9) discussed advances in representation for SLAM, with a particular focus on approaches that go beyond elementary geometry to consider objects, semantics, and hierarchies of spatial relations. W. Chen et al. (10), Tourani et al. (11), and C. Chen et al. (12) covered more recent trends in vision-based SLAM, focusing on the use of deep learning techniques for pose estimation and extraction of semantic information. Placed et al. (13), on the other hand, focused on active SLAM strategies for precise mapping through motion planning while briefly discussing topological, metric, metric-semantic, and hybrid map representations. More recently, Yang et al. (14), Tosi et al. (15), Wang et al. (16), and Ming et al. (17) have provided comprehensive reviews focusing specifically on the latest advancements in radiance field-inspired representations for SLAM and other applications in robotics.

While the most recent literature often narrows its scope to specific advancements in deep learning (12), active SLAM (13), or radiance field methods (14–17), this survey adopts a more holistic approach, categorizing and examining a wide array of scene modeling techniques that have driven progress in SLAM and robotic spatial perception over the past decade. This perspective highlights

the evolution of the field across various dimensions, from online metric scene reconstruction to the integration of semantics and higher-level abstractions, offering a comprehensive understanding of the latest advancements and their interconnections.

3. FOUNDATIONS OF SPATIAL PERCEPTION FOR ROBOTS

Spatial perception is the capability of an embodied device to continuously capture and process sensor data in order to build a persistent, actionable representation of its surrounding environment in real time. Also referred to as spatial AI in a pioneering essay by Davison (18), the term is often employed to refer to the evolution of SLAM into more advanced perception frameworks that intend not only to capture the geometry of the scene but also to integrate higher-level, symbolic knowledge such as semantics, relations, and affordances. Grounding this richer understanding into a spatial representation that can be used for task planning promises to enable more sophisticated and context-aware reasoning, enhancing the robot's ability to navigate, manipulate objects, and collaborate effectively with humans and other autonomous agents in complex, real-world settings.

We start by describing the basic working principles of a spatial perception system, which we illustrate in **Figure 1**. In the most generic case, we consider a sensor suite comprising one or more exteroceptive sensors, such as cameras (RGB or RGB-D) or lidar scanners, and (optionally) supporting proprioceptive sensors, such as inertial measurement units (IMUs). The sensors of choice are typically integrated with a processing unit that is embedded in a robotic platform capable of moving through the world. The goal is to use the raw sensor data to incrementally build a representative map of the robot's surroundings, as well as to estimate the state of the robot relative to this map (such that new sensor measurements can be continuously incorporated into it). Ultimately, the resulting map should provide useful information for the execution of a potentially wide variety of tasks, ranging from safe navigation to intelligent interaction with the scene or even with humans.

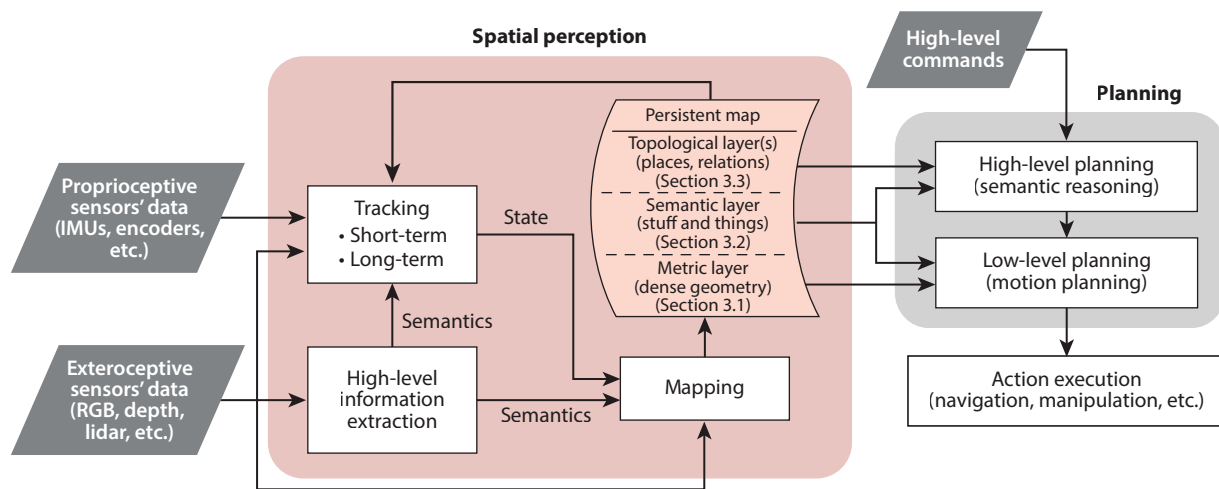


Figure 1

Key components of robotic spatial perception. The main goal of a robotic spatial perception system is to build a persistent and actionable map of the robot's environment from sensor data. To truly enable high-level reasoning, such a map should encode information at different levels of abstraction, namely metric, semantic, and topological. In Section 4, we review key frameworks from the recent literature and categorize them based on the level of abstraction they intend to provide. Abbreviation: IMU, inertial measurement unit.

Broadly speaking, such information can be divided into three different levels of abstraction: metric (i.e., a representation of the scene's geometry), semantic (i.e., identification of objects and other scene entities that have semantic meaning), and topological (i.e., relationships between scene entities, and possibly between those and higher-level spatial abstractions, such as places, which might become particularly useful when operating in large-scale environments). This stems from the observation that, while detailed metric information and object awareness are necessary for motion planning, topological and semantic information is usually better suited to high-level reasoning.

With online operation being imperative in most robotics applications, one of the main challenges in spatial perception is to maintain a sufficiently rich and detailed representation of the environment within strict real-time constraints. Generally, as addressed in SLAM, this is achieved through an iterative process that involves extracting salient information from incoming sensor data, estimating the robot's state based on the newly acquired data and the information already in the map (i.e., tracking), and integrating the new observations into the scene model (i.e., mapping). However, a fundamental question when considering the design of such a system is, What makes a suitable scene representation? While the answer varies depending on the task at hand, the sensors and resources available, and the performance requirements, we outline some key attributes that modern systems need to consider:

- **Metric accuracy:** The representation is often required to accurately capture the geometric details of the environment with high fidelity. Furthermore, spatial consistency should be enforced—i.e., every point in the world should have only one corresponding point in the map. This is essential for tasks that require precise spatial information, such as navigation, obstacle avoidance, and object manipulation.
- **Semantic richness:** Beyond geometry, the representation can incorporate semantic information about the environment. This includes the identification and classification of objects and other scene elements, and possibly additional layers of abstraction, such as rooms (in the case of indoor environments) or places. Semantic richness enables the robot to “understand” the context of its environment, making it possible to execute tasks that require high-level reasoning or interaction with it and with humans.
- **Scalability:** Scene representations should scale from small, confined spaces to larger, complex environments, and even entire cities. In particular, they should allow for fast inference and updates as new sensor data are acquired, and grow in size only when previously unobserved parts of the environment are incorporated. This involves managing memory and computational resources effectively while maintaining high accuracy and detail.
- **Efficiency:** Autonomous systems often operate under stringent real-time constraints, requiring scene representations that allow for quick inference and updates as new sensor information is acquired. This involves the development of algorithms that minimize latency and computational load.
- **Robustness:** Scene representations should be capable of handling sensor noise, varying environmental conditions, and even potential system disruptions, maintaining their functionality and performance across different scenarios. This robustness to commonly expected disturbances ensures that the processes can be trusted to function effectively and safely, even in challenging and unpredictable environments.
- **Long-term persistence:** To enable long-term autonomy, the scene representation needs to be persistent and continuously updated, and therefore capable of accommodating both short- and long-term changes in the environment. This is crucial for tasks requiring ongoing operation in evolving scenarios, such as continuous monitoring, maintenance, or long-duration missions.

- **Multimodal integration:** Robots typically use multiple sensors—such as cameras, lidar, sonar, and tactile sensors, among others—to perceive their environment. A comprehensive scene representation should ideally integrate data from these diverse sensors to create a holistic model of the environment.
- **Interpretability and usability:** For human–robot interaction and collaborative tasks, scene representations need to be interpretable by both robots and humans. This includes intuitive visualizations and interfaces that allow humans to understand and interact with the robot’s perception of the environment.

4. SCENE REPRESENTATIONS

The representation of a scene is the cornerstone of spatial perception, serving as the foundational data structure that allows a robotic system to interpret and interact with its environment, as well as estimate its state relative to it. This section discusses the most prevalent and latest trends in scene representations employed in robotics, analyzing their strengths and limitations. We categorize these representations into three main groups, based on the level of abstraction they provide: metric, metric–semantic, and metric–semantic–topological. **Figure 2** provides an overview of some key SLAM and spatial perception frameworks featured in this review, illustrating their classification within our proposed taxonomy.

4.1. Metric Representations

Metric representations provide information about the geometry of a scene. These low-level representations are crucial for precise ego-motion estimation and constitute the backbone of any

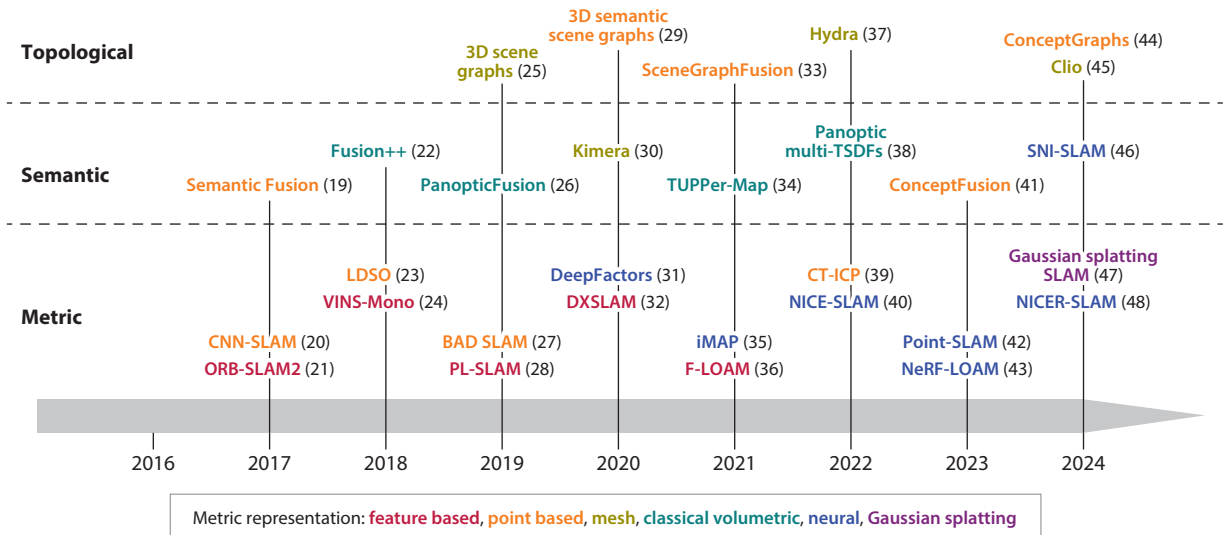


Figure 2

Timeline of scene representations in SLAM and spatial perception frameworks. Key works in the field are listed, sorted by year of publication and the highest level of abstraction (i.e., metric, semantic, or topological) they attempt to provide. The color coding corresponds to the type of metric representation used to model low-level geometry, following the taxonomy introduced in Section 4.1. Since the publication of iMAP (35) in 2021, there has been a notable trend of methods drifting away from traditional representations (i.e., feature-based, point-based, mesh, and classical volumetric representations) toward neural and explicit representations based on radiance fields. Abbreviations: SLAM, simultaneous localization and mapping; TSDF, truncated signed distance field.

SLAM system. However, while some can support basic tasks, such as navigation and obstacle avoidance, they lack semantic context and therefore do not provide the robot with high-level scene-understanding capabilities.

The quest to efficiently and accurately model 3D geometry is a long-standing problem across computer graphics and robotics, and numerous techniques have been developed over the years to tackle this challenge. Here, we review the metric scene representations most commonly used in SLAM and spatial perception systems, with particular emphasis on those developed after 2016 [i.e., following the publication of the survey by Cadena et al. (6)].

In general terms, metric scene representations can be categorized into feature-based representations and dense maps. Feature-based representations only capture a sparse set of key points in the scene and are therefore highly compact, with low computational and memory requirements. This makes them particularly well suited for localization in large-scale settings. However, while efficient, they do not provide fine-grained scene-modeling capabilities and therefore cannot immediately support task planning. Dense representations, on the other hand, seek to provide high-fidelity models of the 3D scene's geometry. Generally, these are more suitable for motion planning and other tasks that require detailed spatial information. To provide a deeper level of specificity, here we subcategorize dense representations into five different types prevalent in the recent literature: dense (and semidense) point-based representations, surface mesh representations, classical volumetric representations, neural implicit representations, and explicit radiance field representations based on 3D Gaussian splatting (3DGS).

4.1.1. Feature-based representations. Feature-based representations aim at capturing a relatively small set of salient, distinctive features—often referred to as landmarks—within an environment. The underlying assumption is that previously mapped features can be reliably recognized in incoming sensor data through data association.

Historically, sparse representations have been widely used in the context of vision-based SLAM. While filtering techniques were the first to be proven capable of building sparse maps—albeit containing only a few tens or hundreds of features (4)—it was soon realized that batch optimization strategies offer a better trade-off between accuracy and computation (49). As a result, subsequent high-performing visual SLAM systems leveraging sparse representations, such as ORB-SLAM (21, 50), have been employing the keyframe-based paradigm introduced in the parallel tracking and mapping method (51), enabling the generation and maintenance of maps with a substantially larger number of landmarks in real time. The inherent sensitivity of purely vision-based frameworks to illumination conditions and motion blur, as well as the problematic estimation of the absolute scale, has motivated substantial research into visual-inertial SLAM systems (24, 52–54), which have proven to be more robust and versatile in a wide variety of environments.

Among the various image features used in computer vision, Shi-Tomasi (55), BRISK (56), and ORB (57) have traditionally been popular choices for visual SLAM algorithms due to their balanced effectiveness, simplicity, and computational efficiency. Recently, with the surge of deep learning, NNs specifically designed for feature extraction have been incorporated within state-of-the-art vision-based SLAM frameworks (32, 58), resulting in increased robustness against environmental and viewpoint changes compared with using hand-crafted features. Beyond point features, a smaller body of research has explored the use of alternative landmarks, such as lines (28, 59) and planes (60), which can offer additional geometric information. Nonetheless, these features have limited applicability and are often more challenging to detect and utilize effectively.

Feature-based representations have also been used for ego-motion estimation in lidar-based SLAM. Prevalent methods, such as LOAM (61) and its variants (36, 62), typically use

geometric features, such as coordinates, distances, normals, curvature, and density, which describe point properties related to their spatial arrangement or geometry.

4.1.2. Dense (and semidense) point-based representations. Among dense models, point-based representations describe the 3D geometry using large, unstructured sets of points. For years, point clouds have been used as the natural map representation produced by SLAM systems working with range sensing modalities, such as RGB-D (63, 64) and lidar (39, 65, 66). Broadly speaking, these methods employ point-cloud registration to compute the most probable pose within the map from which new measurements are captured. Beyond RGB-D and lidar, some classical vision-based SLAM systems have also attempted to build dense (67) or semidense (23, 68, 69) point clouds directly from the raw intensity values of all or a subset of the image pixels, respectively. More recently, combining deep NNs for frame-wise depth prediction with classical optimization back ends (70, 71), as pioneered in CNN-SLAM (20), has shown potential in overcoming previous limitations such as the absolute scale recovery and the problematic depth estimation in textureless areas. Additionally, DROID-SLAM (72) introduced an end-to-end differentiable architecture for updating camera poses and dense per-pixel depth to maximize their compatibility with the current estimate of optical flow.

Another type of scene representation that can be considered within this domain is surfel maps, which consist of disks that capture both the position and the orientation of small surface patches without explicit connectivity. Despite typically requiring the storage of larger amounts of data, surfel maps have also been successfully used in RGB-D-based (27, 73) and lidar-based (74) SLAM.

4.1.3. Surface mesh representations. Surface meshes are composed of sets of polygons (most commonly triangles) that share edges, thus forming a continuous, piecewise planar representation of the observed scene surfaces. In robot mapping, mesh maps are often generated by extracting them on demand from other underlying representations, such as occupancy maps or truncated signed distance fields (TSDFs) (both explained in Section 4.1.4). Real-time methods for building mesh maps incrementally, using pose estimates from a SLAM system, have also been proposed across various sensing modalities, including vision (75), RGB-D (76), and lidar (77). Fewer approaches, however, have attempted to build mesh maps online while enabling localization and meshing to benefit each other. Examples of such integrated systems include the vision-based methods by Rosinol et al. (78) and Bloesch et al. (79), as well as the lidar-based Puma (80) and SLAMesh (81) frameworks.

4.1.4. Classical volumetric representations. Volumetric representations aim to fully capture an observed environment by modeling not only the 3D geometry but also the free space within it. This makes them often preferred over point clouds or meshes for robotic planning and navigation. Historically, the most commonly used scene representations for volumetric mapping have been occupancy maps and signed distance fields (SDFs). The former explicitly represent free and occupied space, while the latter are volumetric fields in which any given point stores the orthogonal distance to the nearest surface, alongside a sign that determines whether that point lies in the interior of the volume whose surface is being estimated.

Classical occupancy-based representations typically employ volumetric grids, where each voxel stores information about whether it is occupied or not. One of the most prominent frameworks in this domain is OctoMap (82), which builds on an octree-based data structure that allows occupancy information to be stored at different resolutions. The success of this approach has led to several follow-up efforts aimed at speeding up map insertions (83, 84). In addition, other grid-based methods have investigated adaptive-resolution strategies to improve efficiency in terms of memory and compute requirements (85, 86).

Despite their practical utility, grid-based sampling approaches might suffer from inaccuracies due to the discrete nature of grids. To address this limitation, some works have adopted the normal distributions transform (87) for modeling occupancy (88, 89). Unlike discrete grid-based methods, approaches based on the normal distributions transform represent each voxel with a local Gaussian distribution, creating a piecewise continuous representation of the environment. As a result, they can achieve comparable levels of modeling accuracy with much larger voxel sizes. In a similar spirit, representations that allow for continuous occupancy mapping, such as those based on Hilbert maps (90) and Gaussian mixture models (91), have also been developed.

Beyond occupancy maps, voxelized TSDFs (i.e., SDF representations that model distances only within a certain threshold value around the surface) have become widely used within online scene reconstruction frameworks due to their ability to achieve subvoxel resolutions and their efficacy in fusing high-rate depth measurements while smoothing out sensor noise. This representation was popularized by the pioneering dense SLAM method KinectFusion (92), although its original, fixed-grid-based approach was soon extended by Nießner et al. (93) to use spatially hashed voxels, allowing for the allocation of new voxels only when needed. To better support trajectory planning, numerous mapping frameworks have aimed at building full SDFs in real time on dynamically growing maps, as these permit efficient queries of distances and gradients to obstacles from any point in space. For instance, following the voxel-hashing approach, Voxblox (94) progressively builds an SDF map from a projective TSDF that integrates the incoming sensor data. In a similar spirit, Voxfield (95) introduced a nonprojective TSDF formulation that results in higher SDF accuracy while maintaining computational efficiency. Moreover, Voxblox's GPU-accelerated extension, nvblox (96), achieves significant speedups in surface reconstruction and further improves the SDF computation. To model expansive maps with varying levels of detail, multiresolution volumetric representations supporting signed distance mapping (97, 98) have also gained significant interest because they are more memory efficient.

4.1.5. Neural implicit representations. In recent years, deep learning models have increasingly been used to encode scene geometry and appearance into implicit representations. Often referred to as neural maps, these representations aim to capture complex spatial information in a more compact and continuous form, thereby addressing the main limitations of the conventional explicit geometric representations discussed in previous sections.

Early works in this domain leveraged deep autoencoders to automatically discover high-level, compact representations of high-dimensional image data, also known as neural latent codes. Their use in visual SLAM was introduced in CodeSLAM (99), which encoded observed images into a compact and optimizable representation containing the essential information of a dense scene. Building on this concept, DeepFactors (31) further integrated this learned implicit representation into a fully featured, keyframe-based SLAM system. The compact nature of the learned representation significantly enhances the efficiency of optimization processes related to camera motion tracking and scene geometry reconstruction, facilitating global consistency in both localization and mapping.

For the past three years, the growing interest in representations derived from neural radiance fields (NeRFs) (100) has significantly influenced the development of new SLAM methodologies based on neural rendering. Leveraging view synthesis as a self-supervision signal to implicitly learn and model the 3D structure and the appearance of a scene, these approaches have been rapidly adopted, especially in the realm of RGB-D SLAM. This new line of research was pioneered in iMAP (35), which uses a multilayer perceptron (MLP) to map 3D coordinates to color and volume density without addressing specularities or considering viewing directions. While this representation enables compact and continuous scene modeling, it often results in oversmoothed

3D surfaces and faces issues such as catastrophic forgetting and slow inference times, limiting the scalability and efficiency of the reconstruction process. To address these shortcomings, approaches like MeSLAM (101) and NISB-Map (102) employ multi-MLP structures to represent different parts of the scene, while numerous other methods exploit grid-based representations. NICE-SLAM (40), for instance, introduced fixed-size, coarse-to-fine feature grids that enable more detailed reconstruction on large indoor scenes. Vox-Fusion (103), on the other hand, leverages an octree-based structure to store grid embeddings, allowing for dynamic allocation of new spatial voxels as the scene expands. Further developments include Co-SLAM (104), which combines the smoothness provided by one-blob encodings (105) with the fast convergence and local detail advantages of a multiresolution hash-based feature grid (106), and ESLAM (107), which replaces feature grids with multiscale, axis-aligned feature planes, reducing the growth of scene scale from cubic to quadratic. Beyond grid-based methods, frameworks leveraging hybrid neural point-based representations, such as Point-SLAM (42) and Loopy-SLAM (108), have recently been developed. Unlike grids, point densities in these representations adapt naturally without the need for prespecified resolutions, leading to more efficient memory use and a greater ability to capture fine details. As a result, point-based methods can achieve superior performance in 3D reconstruction.

The success of NeRF-style approaches in RGB-D SLAM has also motivated substantial research into methods tailored to RGB and lidar sensing. In the domain of RGB SLAM, iMODE (109) uses ORB-SLAM2 (21) for real-time camera pose estimation and concurrently optimizes an MLP representing the neural field. Similarly, both Orbeez-SLAM (110) and NeRF-SLAM (111) integrate well-established visual odometry modules [ORB-SLAM2 (21) and DROID-SLAM (72), respectively] for camera tracking with hierarchical volumetric NeRF modeling based on instant neural graphics primitives (106), achieving real-time operational efficiency. NICER-SLAM (48), on the other hand, introduced a unified end-to-end framework that simultaneously optimizes for camera poses and a hierarchical neural implicit map representation modeling SDF and color information. Among lidar-based strategies, NeRF-LOAM (43) jointly estimates the six-degrees-of-freedom sensor pose while constructing a neural representation that employs dynamic voxel embeddings within an octree-based architecture, effectively capturing local geometry. LONER (112), by contrast, employs the iterative closest point method for odometry estimation and represents the scene by using an MLP with a hierarchical feature grid encoding, while PIN-SLAM (113) leverages a point-based implicit neural map representation in combination with correspondence-free point-to-implicit model registration for pose estimation, allowing for continuous deformation during global pose adjustments.

4.1.6. Explicit radiance field representations. The main drawback of NeRF-style approaches is that the volumetric ray-sampling process required for rendering significantly limits their efficiency. Promising alternatives include explicit radiance field representations based on the novel 3DGS paradigm (114), which models 3D scenes in terms of differentiable 3D Gaussian-shaped primitives, as shown in **Figure 3**, and achieves state-of-the-art rendering accuracy while boosting rendering speed. The pioneering frameworks that leverage scene representations based on 3DGS for SLAM are GS-SLAM (115) and SplatAM (116) in the context of RGB-D sensing, Gaussian splatting SLAM (47) for monocular SLAM, and LIV-GaussMap (117), which is designed for lidar-inertial-vision fusion. Despite their growing popularity, these methods face several challenges, such as a significant dependence on initialization and limited control over the growth of primitives in unobserved regions. Additionally, the original 3DGS-based scene representation necessitates a large number of 3D Gaussian primitives to achieve high-fidelity reconstruction, leading to substantial memory consumption.

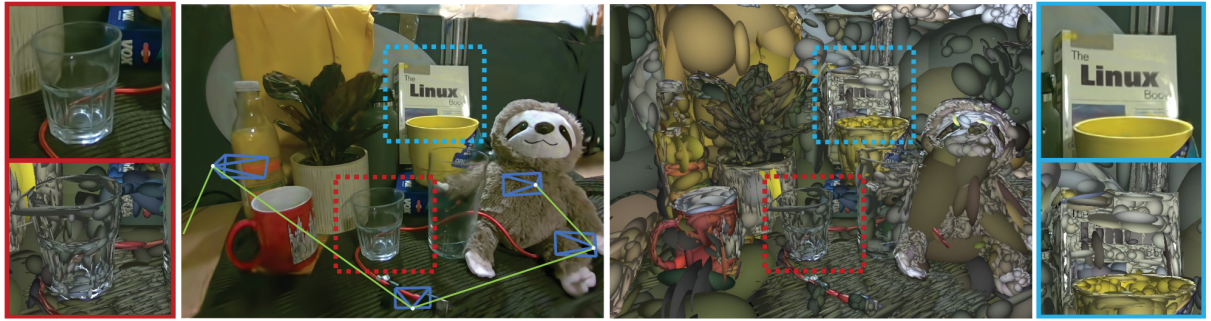


Figure 3

Visualization of a scene representation based on 3D Gaussian splatting. The large image on the left shows the rasterized Gaussians, and the one on the right shows the shaded Gaussians to highlight the resulting map geometry; the smaller images on the outside show close-ups of two specific parts of the scene. It is worth noting how the sizes and concentration of such Gaussians adapt to the complexity of each part of the scene being modeled, leading to high image-rendering quality. Figure adapted with permission from Reference 47; copyright 2024 IEEE.

4.2. Metric–Semantic Representations

To enable high-level reasoning or to aid physical interaction tasks, modeling geometry alone is not sufficient. In other words, the reconstructed maps must encode information regarding semantically meaningful scene entities, such as objects, providing robots with a deeper understanding of the content and the structure of the surrounding space. For example, identifying a flat surface as a table, rather than just a generic plane, can facilitate making appropriate decisions about placing certain objects on it.

Aiming at addressing these needs, metric–semantic representations go beyond purely metric representations by attaching semantic labels (e.g., object categories) to them. Interest in this dual-layered approach has grown significantly since the surge of deep learning, which in its early days already demonstrated remarkable capabilities in object detection, classification, and scene understanding. Nowadays, state-of-the-art deep learning models are commonly used to process the vast amounts of incoming sensor data and identify objects and other semantically meaningful entities within an environment, providing the necessary high-level information to build metric–semantic representations. Methods in this field can be divided into four groups, depending on whether they intend to build object-centric, semantically labeled, panoptic, or open-set representations.

4.2.1. Object-centric representations. Methods based on the object-oriented approach focus on identifying and reconstructing a set of objects of interest, while typically ignoring the semantics of the rest of the observed scene. Early works on online object-oriented mapping (118) leveraged 3D model databases, requiring the shapes of the objects in the scene to be exactly the same as the prelearned models and therefore inapplicable to real environments, where objects with geometric variations are frequently encountered. With the surge of deep learning, several methods adopted convolutional neural network architectures for detecting objects in RGB images. In this context, efficient object representations were developed to enable the estimation of the class, pose, and size of objects within landmark-based monocular SLAM frameworks. These frameworks include the dual quadric formulation, which models objects as 3D ellipsoids (119, 120), and CubeSLAM (121), which represents objects in terms of rectangular bounding volumes. Concurrently, methods tailored to RGB-D sensing aimed at densely reconstructing the environment, while accurately segmenting the recognized objects in it. Sünderhauf et al. (122), for example, combined an unsupervised geometric segmentation scheme with a learning-based 2D bounding-box

object detector (123) to identify and merge geometric 3D segments being recognized as part of the same instance. Fusion++ (22), on the other hand, leveraged Mask R-CNN (124), which predicts semantically annotated 2D segmentation masks for the objects recognized in the input images, to progressively build individual 3D object models that were used as landmarks in an optimizable pose graph. Following this line of research, Voxblox++ (125) combined Mask R-CNN with an incremental geometric scene segmentation approach (126) to produce a complete instance-aware semantic mapping framework, aiming to retrieve the pose and shape of both recognized objects and newly discovered, previously unobserved object-like instances. Later, Mascaro et al. (127) used the Mask R-CNN predictions to extract 3D instance segments that were further refined after being integrated into a global dense map using a label diffusion scheme, leading to a flexible framework tailored to both RGB-D and vision–lidar sensing.

While the frameworks mentioned above used point clouds (122) or TSDF maps (22, 125, 127) to represent geometry, more recently, vMAP (128) has introduced a novel approach to object-level dense SLAM by representing each object with a dedicated MLP. This strategy facilitates the creation of watertight and complete object models, even when dealing with partially observed or occluded objects in real-time RGB-D input streams. In a similar spirit, RO-MAP (129) combines a lightweight object-centric SLAM with NeRF models for simultaneous localization and reconstruction of objects from monocular RGB input. The system efficiently trains separate NeRF models for each object, demonstrating real-time performance in semantic object mapping and shape reconstruction without depth priors.

4.2.2. Semantically labeled representations. Semantic labeling approaches aim at assigning a class label or a probability distribution of class labels to each point, surfel, or voxel in the reconstructed 3D map. One of the most representative frameworks following this paradigm is the RGB-D-based SemanticFusion (19), which for the first time leveraged a convolutional neural network to infer per-pixel class probability distributions that were aggregated onto a surfel-based map using a Bayesian update scheme. Building on similar ideas, Kimera (30) introduced a real-time metric–semantic SLAM system that integrates frame-wise semantic segmentation, IMU measurements, and optional depth measurements into a dense mesh of the environment, while SuMa++ (130) and SA-LOAM (131) were proven capable of building semantically annotated maps using lidar input.

More recently, novel RGB-D-based approaches leveraging radiance field representations have been developed. SNI-SLAM (46) employs a neural implicit representation, hierarchical semantic encoding for multilevel scene comprehension, and cross-attention mechanisms for collaborative integration of appearance, geometry, and semantic features. SGS-SLAM (132) and SemGauss-SLAM (133), on the other hand, are the first semantic dense SLAM systems built on the 3DGS paradigm. The former embeds semantic information in the 3D Gaussians as additional color channels that are optimized over 2D semantic segmentation maps predicted on the RGB keyframes, while the latter extends the 3D Gaussians with semantic feature embeddings, achieving higher 3D semantic segmentation accuracy.

4.2.3. Panoptic representations. Despite allowing for higher-level scene understanding, semantic labeling approaches lack the ability to distinguish individual instances belonging to the same category, and as a result, information about the number, geometry, and relative placement of individual objects in the scene is limited. To address this issue, the so-called panoptic representations seek to incorporate class labels of background regions (stuff) while individually segmenting and recognizing arbitrary foreground objects (things), as illustrated in **Figure 4**.

Before the term panoptic was even coined (135), researchers had already attempted to solve this problem. For instance, Nakajima et al. (136) combined per-frame semantic segmentation

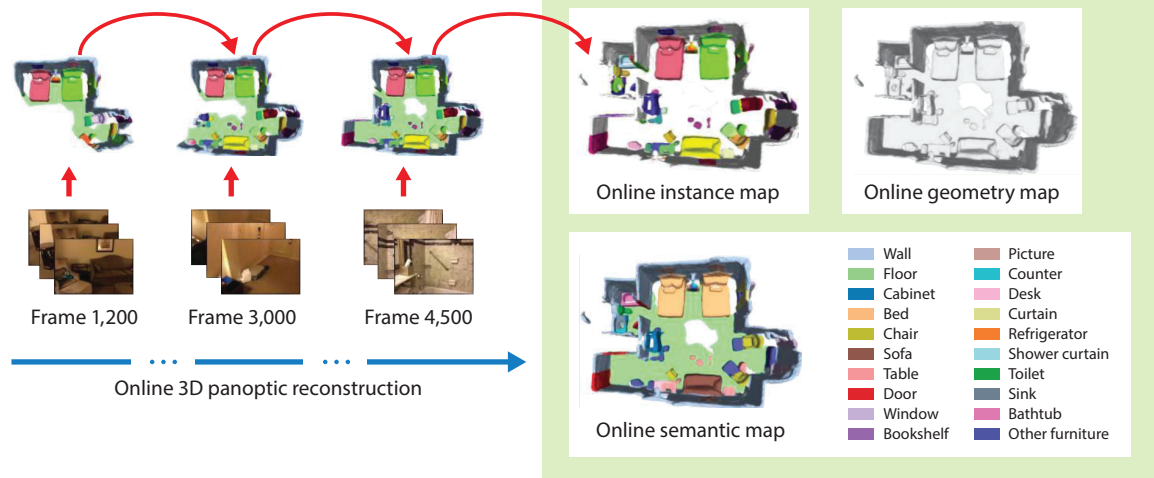


Figure 4

Online 3D panoptic reconstruction. Panoptic mapping frameworks perform incremental dense 3D geometry reconstruction (geometry map), semantic labeling for all map regions (semantic map), and segmentation of individual objects (instance map). The figure shows sample input data and the 3D maps generated by PanoRecon (134) in an indoor scene. Image adapted with permission from Reference 134; copyright 2024 IEEE.

with an incremental scene segmentation approach using geometric cues from depth, while Pham et al. (137) employed a purely semantic segmentation strategy and then clustered the semantically annotated scene into individual instances. Without instance-level semantic information as input, however, geometry-based approaches tend to oversegment articulated scene elements, thus failing to model individual object instances accurately. More recent RGB-D-based mapping approaches (26, 34, 138) explore the deep-learning-based panoptic segmentation paradigm (135) in combination with classical volumetric representations. Furthermore, panoptic multi-TSDFs (38) have been introduced for adaptive multiresolution volumetric mapping. This approach models the world as a collection of submaps, each representing an object instance or a piece of background with its own required resolution.

Going beyond classical volumetric representations, PanoRecon (134) proposes a novel framework for real-time, incremental joint 3D geometric reconstruction and 3D panoptic segmentation in a view-independent 3D feature volume, given a posed monocular video. In addition, NeRF representations decomposing the scene into a set of objects and background have also been proposed (139, 140). However, these rely on time-consuming per-scene optimization and have only been designed to run offline so far.

4.2.4. Open-set representations. The vast majority of existing approaches that incorporate semantic concepts into 3D maps remain largely confined to a closed-set paradigm, where only a predefined, finite set of objects or categories, established during training, can be recognized. However, real-world environments are inherently unpredictable and open-set, often presenting robots with novel objects or scenes that may fall outside of their predefined knowledge base. To address this challenge, advancements in open-set representations driven by foundation models have recently emerged. These novel frameworks enable the fusion of open-set features into 3D maps, allowing for flexible querying and interpretation later on. One of the most representative examples of this emerging field is ConceptFusion (41), which demonstrates that

pixel-aligned open-set features can be fused into 3D maps via traditional SLAM and multiview fusion approaches, enabling effective zero-shot spatial reasoning.

4.3. Metric–Semantic–Topological Representations

While low-level motion planning and obstacle avoidance typically rely on fine-grained map representations, such as meshes or volumetric models, high-level task planning focuses on outlining a series of actions to accomplish a specific goal. Connecting low-level and high-level planning, especially when it comes to understanding and executing human instructions such as “navigate to the kitchen and pick the coffee mug I left on the counter,” requires building an actionable world model that is aware of the relationships between semantic entities and encapsulates reality at different levels of abstraction. To address this problem, recent work has attempted to design richer scene representations, called 3D scene graphs, containing metric, semantic, and topological information. In general terms, 3D scene graphs are graphical models in which nodes represent spatial or semantic concepts and edges represent relations between those concepts. Such models can be flat, if they solely focus on relationships among concepts at a single level of abstraction (e.g., objects), or hierarchical, if they model semantic and spatial concepts at multiple levels of abstraction (i.e., from low-level geometry to objects, places, rooms, buildings, etc.).

The use of 3D scene graphs was pioneered by Armeni et al. (25) and Kim et al. (141). The former introduced a semiautomatic algorithm to build a hierarchical 3D scene graph model containing objects and rooms from a metric–semantic 3D mesh, while the latter designed a method to reconstruct flat 3D scene graphs only including objects and their relations. Subsequent work on flat 3D scene graphs has further explored learning-based approaches to infer objects and spatial relationships among them from class-agnostic, clustered point clouds (29) (offline) and from RGB-D sequences (33) (incrementally). Hydra (37, 142), on the other hand, has introduced a hierarchical 3D scene graph model that can be built directly from visual–inertial data in real time. This novel approach reconstructs a local Euclidean SDF of the robot’s surroundings and incrementally converts it into a metric–semantic 3D mesh, from which a topological graph of objects, places, and rooms can be extracted, as shown in **Figure 5**. More recently, methods leveraging foundation models for building flat (44) and hierarchical (45) open-set 3D scene graphs have also started to emerge.

4.4. Dynamic Scene Representations

Historically, the immense majority of SLAM and spatial perception frameworks have assumed that the world is static. Real-world environments, however, are often dynamic, featuring moving humans and objects, continually evolving conditions (e.g., illumination and weather), and changing landscapes. Therefore, providing robots with the ability to perceive and adapt to both short- and long-term changes in the environment is crucial for effective operation. This section explores various approaches and techniques that have recently emerged to tackle scene dynamicity in the realm of robotic perception.

4.4.1. Short-term dynamics. Many research efforts in the past few years have focused on addressing directly observable motion. In particular, feature-based SLAM methods have focused on maintaining state estimation robustness, typically by filtering out dynamic entities (143–145) or integrating some of those entities into the SLAM problem, assuming the availability of good motion priors (146–149). A family of RGB-D SLAM systems (150–152), on the other hand, have leveraged learned object detectors to simultaneously track and densely reconstruct multiple rigid objects moving within the sensor’s field of view. Ren et al. (153) have further introduced a re-localization method to recover moving objects that disappear and reappear in the camera view.

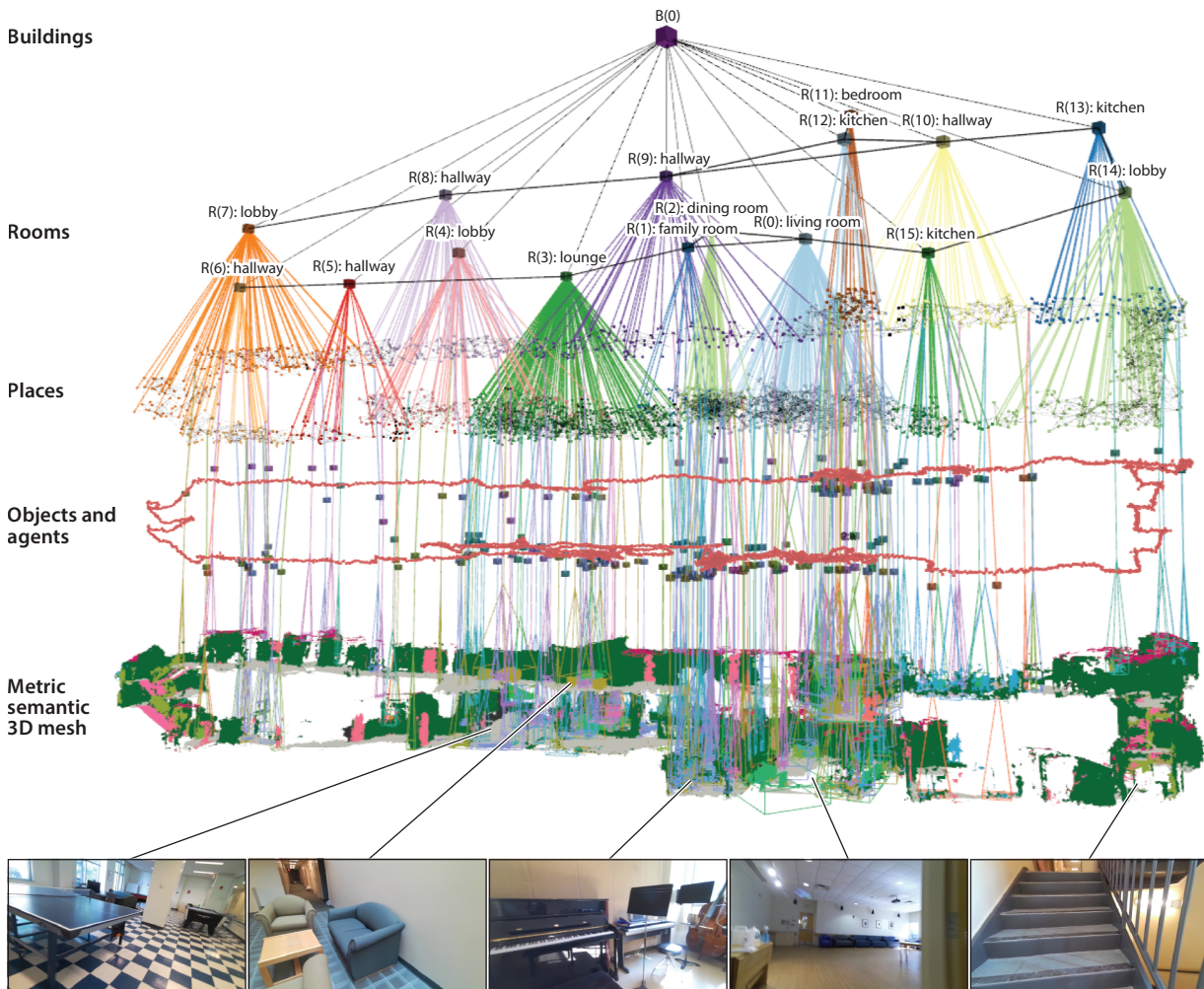


Figure 5

A hierarchical 3D scene graph. Hierarchical 3D scene graphs are scene representations that abstract a dense, often semantically labeled 3D model into higher-level spatial concepts, from objects to places (i.e., clusters of free space). In the case of indoor environments, places can be further abstracted into rooms and even buildings. The figure shows sample input data and the hierarchical 3D scene graph generated by Hydra (142) in a large-scale real environment. Figure adapted with permission from Reference 142 (CC BY-NC 4.0).

In a similar spirit, some recent lidar-based approaches (154, 155) have also proven capable of performing joint dynamic foreground and static background reconstruction in larger scenes.

4.4.2. Long-term dynamics. An important drawback of the aforementioned methods is that they are not designed to identify long-term changes (i.e., those that become observable only when revisiting a certain location over time) inherent to semistatic scenes. While the problem of change detection is not new, it has only recently started to be tackled from the perspective of an online consistent mapping setting. Assuming known robot poses, initial attempts have managed to handle changes by evaluating the consistency of object-level (156) and panoptic (38) TSDF submaps, as well as neural object models (157). More recently, POV-SLAM (158) has

introduced an object-aware, factor-graph SLAM formulation capable of tracking and reconstructing semistatic object-level changes.

4.4.3. Short- and long-term dynamics. Fewer works have explicitly addressed the problem of combining short- and long-term dynamics in a single framework. Initial attempts are represented by Changing-SLAM (159), which builds a map of sparse key points, employs a Bayesian filtering approach to detect short-term dynamics and data association to detect long-term changes, and removes any detected dynamic points from the SLAM problem. More recently, Schmid et al. (160) introduced Khronos, the first framework capable of generating a spatiotemporal metric-semantic map in real time, jointly optimizing for the robot poses and an explicit dense semantic representation of the scene at all times.

4.4.4. Deformable environments. Another problem that is particularly challenging and rather underexplored is that of mapping deformable environments, or environments containing non-rigidly deformable objects. A pioneering work in this domain was DynamicFusion (161), which demonstrated the dense reconstruction of deformable objects using an RGB-D camera. More recently, MORPH-DSLAM (162) applied computational mechanics as a tool for estimating the deformations exhibited by objects perceived with a standard camera. In addition, other approaches have explored the problem of SLAM in deformable environments using monocular (163, 164) and stereo (165) cameras, with particular focus on medical endoscopy applications.

5. DISCUSSION

The past decade has witnessed remarkable advancements in the representational capabilities of modern robotic perception systems. However, a number of fundamental issues persist. In this section, we discuss these open challenges and highlight promising avenues for future research in the field.

5.1. A Universal Scene Representation

While it is generally acknowledged that advanced spatial perception will be a critical enabler of the next generation of robots and autonomous systems, it is not yet clear what the underlying scene representation should be. The challenge lies in identifying an optimal scene representation that can strike a balance among several competing attributes: accuracy, efficiency, scalability, and interpretability. These factors are crucial for meeting the diverse requirements of autonomous systems intended to operate intelligently in the real world.

The broad range of tasks that robots are expected to perform and the diversity of environments that need to be considered for potential deployment translate into a growing variety of platforms and sensors being used. At the same time, the choice of a suitable scene representation is strongly tied to the specific requirements of the task at hand and the available sensors. Each task may demand a different level of detail, semantic understanding, or temporal persistence from the map, while the capabilities and limitations of the onboard sensors dictate the type and quality of information that can be captured. As a result, it is unrealistic to expect that a single, universal representation will suffice for handling all possible use cases. Nonetheless, despite clear differences in the way 3D geometry is modeled in systems today, there seems to be a clear need to abstractify geometry into higher-level concepts, allowing for semantic-based reasoning. Therefore, there is great potential in developing general, flexible frameworks capable of integrating task-relevant information at different levels of abstraction (i.e., from low-level geometric details to objects, entities, and places) that can be queried in a somewhat standardized fashion. How these frameworks can be effectively structured to accommodate various metric representations, semantic abstractions, and temporal scales is still an open problem.

5.2. Real-Time and Scalability Considerations for Practical Deployment

One of the main limitations of current spatial perception frameworks, especially those that aim to go beyond purely geometric SLAM, is their difficulty in achieving real-time processing. This is often caused by the computational complexity of the chosen map data structures and the methods used to update the information contained in them. Moreover, many of these systems, such as those building upon novel neural representations, require powerful GPU hardware, which prevents their deployment on resource-constrained platforms. A critical challenge moving forward is the development of faster, more efficient algorithms capable of running on embedded systems with limited computational power, enabling wider application of advanced perception frameworks.

Another unresolved issue is the need to ensure that spatial perception systems can scale to large and complex environments while maintaining accuracy. The investigation of metric representations that can automatically adapt their level of detail depending on the complexity of the environment, such as multiresolution volumetric maps or 3DGS-based representations, seems a promising step in that direction. Furthermore, when operating in relatively large environments, it may not be necessary for a robot to have instant access to all detailed metric information at all times. This observation suggests the potential for developing advanced models that can selectively retrieve relevant data from long-term memory into short-term memory, based on the robot's current location and task requirements. Such a concept opens up an intriguing area for further research, as it promises to reduce the robot's cognitive load and optimize processing efficiency.

5.3. View-Based Versus Map-Based Inference of High-Level Concepts

The prevailing approach to incorporating semantics into spatial perception systems is to process the input sensor data through learned perception models, treating their output as virtual sensor measurements that can be incrementally fused into the robot's model of the world. The main reason for adopting this view-based approach, as opposed to inferring high-level concepts from the map itself, is the higher fidelity of raw sensor data (e.g., images) compared with a compressed, potentially noisy scene representation, as well as the fact that NNs for processing raw sensor data are much more mature and benefit from a wider availability of training data compared with map-based NNs, often leading to better generalization capabilities. Another key advantage of view-based approaches lies in their ability to provide immediate access to high-level concepts during online mapping, bypassing the need for a partial or complete metric 3D reconstruction of the scene inherent to map-based approaches.

Although map-based approaches have not been discussed in this article because they are designed primarily for offline purposes, it is worth pointing out that there is also substantial work on the topic within the computer vision literature. So far, however, little effort has been put into deriving semantics from a growing map or the history of a robot's interactions with its environment and with humans. This gap in research highlights an opportunity to develop more sophisticated systems that can dynamically update their semantic understanding of a scene as they observe and interact with their surroundings over time. Incorporating temporal, contextual, and interactional data could provide deeper insights, enhancing the robot's overall spatial awareness. In this context, exploring hybrid methods combining both view-based and map-based inference of semantics seems to be an interesting avenue for further research. Such systems would leverage the real-time adaptability and responsiveness of view-based methods alongside the better contextual reasoning provided by map-based approaches. This combination holds the potential to significantly enhance the robustness and reliability of the robot's semantic perception, enabling it to operate more effectively in diverse environments.

5.4. Toward Long-Term Autonomy

Current state-of-the-art spatial perception frameworks have mainly been demonstrated under the assumption of relatively accurate pose estimates, as well as spatially and/or temporally bounded deployment. However, the need for these systems to operate over extended periods and build persistent scene representations raises several challenges.

First, the closed-loop nature of spatial perception systems requires efficient global optimization methods. Current loop-closure and global bundle adjustment algorithms typically take up significant computational resources, especially in real-time applications, and the complexity of updating an entire 3D model often becomes prohibitive. While approaches based on submapping have been proposed to mitigate this issue, the complexity of optimizing an ever-growing map remains an open problem.

Another issue inherent to long-term autonomy is that environments are usually subject to changes over time. Consequently, exploring representational approaches that can effectively detect and adapt to these changes is fundamental. In particular, an important novel challenge in this context is the need to explicitly account for, and actively manage, the uncertainty introduced by an ever-changing world.

Finally, long-term operation will also require robots to embrace continual learning and adaptation. These systems need to operate under open-set conditions, as they will inevitably encounter novel objects and scenarios through continuous observation. To date, however, most NNs used in robotic perception are trained on closed-world settings with a fixed number of semantic categories. Although open-set approaches leveraging foundation models have started to emerge recently (see Section 4.2.4), a significant challenge moving forward is to harness the power of deep networks in a lifelong learning framework, enabling the inference of new semantic concepts and improving their adaptability to previously unseen environments.

5.5. Evaluation: Benchmarking Datasets and Real-World Experiments

One final question is how to properly evaluate the effectiveness of a spatial perception system. Traditionally, the primary performance metric in SLAM has been pose estimation accuracy, and significant efforts have been made to create benchmarking datasets for this purpose. However, spatial perception encompasses more than just pose estimation, and other performance metrics of general importance, such as dense reconstruction accuracy, object segmentation and classification accuracy, system latency, and memory usage, among others (18), should be taken into consideration.

While some attempts to create real-world datasets containing raw sensor sequences together with dense, semantically annotated scene models have already been made, these have been limited mainly to indoor scenes and confined urban areas. The lack of larger and more varied datasets, tailored to the problem of SLAM and spatial perception, hinders the evaluation of current frameworks in terms of robustness, generalizability, and long-term performance. Unfortunately, collecting high-quality datasets for that purpose is extremely costly and time-consuming. An alternative might be to use synthetic data generated using computer graphics, which can provide ground-truth geometry and semantics with greater ease and accuracy. As simulation technology gets better and more realistic, it will become an even more valuable tool for thorough quantitative analysis, especially for complex use cases involving large-scale and dynamic environments. In addition, with the field becoming more populated with mature and sophisticated approaches to the problem of spatial perception under various conditions, the need for simulators and general consistency among evaluation protocols will become increasingly evident. Nonetheless, it is equally important to encourage the evaluation of methods on real hardware and in real settings, as it is these insights from reality that will continue to push progress toward practical systems.

6. CONCLUSION

Inspired by recent advances pushing scene mapping toward scene understanding with the goal of boosting the autonomous capabilities of robots, this survey has reviewed the landscape of scene representations for robotic spatial perception, resulting in a comprehensive overview that traces the historical evolution of the field and, most importantly, highlights some of the most promising trends at present. Significant emphasis has been placed on the transition from purely geometric models to more advanced representations that incorporate semantic and topological information. This evolution marks a crucial step forward in the development of intelligent robotic systems, as these more sophisticated representations hold the potential to dramatically enhance the high-level reasoning and decision-making capabilities of robots.

Although the field of robotic spatial perception has made significant strides over the past decade, it continues to face exciting challenges and opportunities. While no single scene representation seems to be universally suitable across all tasks and application areas, the research focus turns toward more complex and information-rich models that promise to be useful in enhancing robustness and adaptability, which are common bottlenecks in today's robotic systems. The efficiency, scalability, and robustness of such frameworks remain largely the open questions that the community is called to address in the coming years in the quest to enable autonomy in increasingly complex and dynamic environments.

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